

Open-Economy Production Network New Keynesian Model

Model Equations, Calibration, and Dynare Implementation

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1 Model Overview

This model extends ? to a small open economy. The closed-economy model shows that, with multiple sectors and intermediate-input linkages, productivity shocks generate an inflation–output tradeoff for every standard price index except a unique *divine coincidence* (DC) index, which weights sectors by Domar weights scaled by price stickiness. We add three open-economy ingredients: (i) imported intermediate inputs (with home/foreign Armington nesting

at each production node), (ii) imported final consumption goods, and (iii) a single internationally traded bond with a debt-elastic premium.

The key result is that the exchange rate enters sectoral marginal costs exactly like a productivity shock scaled by the import-content matrix $\mathbf{\Omega}^F$. By embedding import nodes as fully flexible *upstream sectors* ($\widehat{\delta}_M = 1$) in an augmented network, the DC index still exists and is immune to both domestic TFP shocks and FX/import-price shocks. A float targeting the DC index implements zero output gap and fully insulates the economy from external shocks via exchange-rate adjustment. A hard peg forces the full adjustment onto domestic inflation and output.

The model is calibrated to a **three-sector oil-exporting economy**: an upstream Resource sector (import-intensive inputs, most output exported), a Manufacturing sector (uses resources as intermediates), and a Services sector (sells mainly to domestic consumers). Three shocks are studied—an import price shock (ε^{pF}), a domestic TFP shock (ε^a), and a foreign demand shock (ε^D)—under three policy regimes: free float, hard peg, and managed float.

2 Environment

2.1 Indexing and notation

There are $N = 3$ domestic sectors indexed $i, j \in \{1, 2, 3\}$ (Resource, Manufacturing, Services). Lower-case hats denote log-deviations from the non-stochastic steady state; tildes denote gaps (sticky-price minus flex-price). $E_t[\cdot]$ is the rational expectation conditional on information at t .

Network matrices.

$$\begin{aligned}\Omega_{ij}^H &= \text{domestic cost share of input } j \text{ in sector } i\text{'s costs,} \\ \Omega_{ij}^F &= \text{imported cost share of input } j \text{ in sector } i\text{'s costs,} \\ \alpha_i &= 1 - \sum_j (\Omega_{ij}^H + \Omega_{ij}^F) = \text{labour share in sector } i.\end{aligned}$$

The domestic Leontief inverse is $\mathbf{L}^H \equiv (\mathbf{I} - \mathbf{\Omega}^H)^{-1}$. The **domestic supplier centrality** (open-economy Domar weight) is

$$\lambda_{D,i} \equiv [\boldsymbol{\beta}^{H\top} \mathbf{L}^H]_i, \quad (1)$$

where $\boldsymbol{\beta}^H = (\beta_1^H, \beta_2^H, \beta_3^H)^\top$ are domestic final consumption shares.

Calvo adjustment. δ_i is the probability that a firm in sector i resets its price each period. Following ?, define the adjusted parameter

$$\widehat{\delta}_i \equiv \frac{\delta_i(1 - \beta(1 - \delta_i))}{1 - \beta\delta_i(1 - \delta_i)}. \quad (2)$$

2.2 Households

The representative household maximises

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]. \quad (3)$$

Aggregate consumption is a CES composite of a domestic bundle C_t^H and an imported bundle C_t^F :

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad (4)$$

where ω is home bias in final consumption and η is the home-foreign elasticity. The domestic bundle aggregates across sectors with shares β^H ; the foreign bundle aggregates across M imported goods with shares μ and elasticity σ .

The budget constraint is

$$P_t^C C_t + \frac{B_{t+1}}{1+i_t} + S_t \frac{B_{t+1}^*}{1+i_t^{*,\text{eff}}} \leq W_t L_t + \Pi_t - T_t + B_t + S_t B_t^*, \quad (5)$$

where S_t is the nominal exchange rate (home currency per unit of foreign), B_t is a domestic bond, B_t^* is a foreign-currency bond, and Π_t are firm profits rebated lump sum.

The effective world rate includes a **debt-elastic premium**:

$$i_t^{*,\text{eff}} = i^* - \psi \left(\frac{S_t B_t^*}{P_t^C} - \bar{b}^* \right), \quad \psi > 0, \quad (6)$$

which ensures stationarity of net foreign assets (requires $\psi > 1/\beta - 1$).

2.3 Firms

Each domestic sector i has a unit mass of monopolistically competitive firms with CRS production:

$$Y_{ift} = A_{it} F_i(L_{ift}, \{X_{ijft}\}_{j=1}^N). \quad (7)$$

The intermediate input bundle X_{ijft} is an Armington composite of domestic and imported varieties of good j :

$$X_{ijft} = \left[v_{ij}^{1/\xi} (X_{ijft}^H)^{\frac{\xi-1}{\xi}} + (1-v_{ij})^{1/\xi} (X_{ijft}^F)^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}}. \quad (8)$$

Cost minimisation gives nominal marginal cost

$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j \tilde{P}_{ijt}^{\omega_{ij}}}{A_{it}}, \quad (9)$$

where $\tilde{P}_{ijt}$ is the Armington price of the input bundle (weighted average of domestic price P_{jt} and foreign price $S_t P_{F,jt}^*$). Firms face Calvo pricing: fraction δ_i reset optimally each period.

2.4 Policy and shocks

The central bank sets the nominal interest rate i_t under one of three regimes (Section 3.10). Exogenous shocks are AR(1) processes.

3 Log-Linearised System

All variables are log-deviations from the zero-inflation steady state. The full system for the three-sector model is listed below; these are the equations entered directly into Dynare.

3.1 Sectoral Phillips curves

For each sector $i \in \{1, 2, 3\}$, the Calvo pricing condition yields:

$$\pi_{it} = \beta(1 - \hat{\delta}_i) \mathbb{E}_t \pi_{i,t+1} + \hat{\delta}_i \tilde{m}c_{it} \quad (10)$$

where the **real marginal cost gap** is

$$\tilde{m}c_{it} = \alpha_i \hat{w}_t + \sum_j \omega_{ij}^H p_{jt} + \omega_i^F (\hat{s}_t + \hat{p}_t^{F*}) - \hat{a}_{it} - p_{i,t-1} \quad (11)$$

with $\omega_i^F \equiv \sum_j \Omega_{ij}^F$ (sector i 's total import share in costs) and $p_{jt} = p_{j,t-1} + \pi_{jt}$.

Sector 1 — Resource: no domestic inputs ($\Omega_{1j}^H = 0$), 30% imported machinery.

$$\pi_{1t} = \beta(1 - \hat{\delta}_1)\mathbb{E}_t\pi_{1,t+1} + \hat{\delta}_1[\alpha_1\hat{w}_t + \omega_1^F(\hat{s}_t + \hat{p}_t^{F*}) - \hat{a}_{1t} - p_{1,t-1}] \quad (12)$$

Sector 2 — Manufacturing: 20% from domestic Resource, 10% imported.

$$\pi_{2t} = \beta(1 - \hat{\delta}_2)\mathbb{E}_t\pi_{2,t+1} + \hat{\delta}_2[\alpha_2\hat{w}_t + \omega_{21}^H p_{1t} + \omega_2^F(\hat{s}_t + \hat{p}_t^{F*}) - \hat{a}_{2t} - p_{2,t-1}] \quad (13)$$

Sector 3 — Services: 25% from domestic Manufacturing, 5% imported.

$$\pi_{3t} = \beta(1 - \hat{\delta}_3)\mathbb{E}_t\pi_{3,t+1} + \hat{\delta}_3[\alpha_3\hat{w}_t + \omega_{32}^H p_{2t} + \omega_3^F(\hat{s}_t + \hat{p}_t^{F*}) - \hat{a}_{3t} - p_{3,t-1}] \quad (14)$$

3.2 Price level laws of motion

$$p_{it} = p_{i,t-1} + \pi_{it}, \quad i = 1, 2, 3. \quad (15)$$

3.3 Real wage (labour supply)

Log-linearising the intratemporal optimality condition and using Lemma 2 (output gap = employment gap):

$$\hat{w}_t = \hat{p}_t^C + (\gamma + \varphi)\tilde{y}_t, \quad (16)$$

where the **consumer price level** in log-deviations is

$$\hat{p}_t^C = \sum_i \beta_i^H p_{it} + \beta^{F,\text{tot}}(\hat{s}_t + \hat{p}_t^{F*}). \quad (17)$$

3.4 CPI inflation

$$\pi_t^C = \sum_i \beta_i^H \pi_{it} + \beta^{F,\text{tot}}[(\hat{s}_t + \hat{p}_t^{F*}) - (\hat{s}_{t-1} + \hat{p}_{t-1}^{F*})]. \quad (18)$$

3.5 Divine coincidence (DC) index

$$\pi_t^{DC} = \sum_i w_i \pi_{it}, \quad w_i = \frac{\lambda_{D,i}(1 - \hat{\delta}_i)/\hat{\delta}_i}{\sum_k \lambda_{D,k}(1 - \hat{\delta}_k)/\hat{\delta}_k}. \quad (19)$$

By Proposition 1, $\pi_t^{DC} = \rho\mathbb{E}_t\pi_{t+1}^{DC} + \kappa^{DC}\tilde{y}_t$ with no residual term.

3.6 IS curve

Log-linearising the Euler equation and imposing market clearing:

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}(\hat{r}_t - \mathbb{E}_t\pi_{t+1}^C - r_{t+1}^{nat}) + \nu_D\hat{D}_t^*, \quad (20)$$

where r_{t+1}^{nat} is the natural (flex-price) real interest rate and $\nu_D > 0$ captures the direct effect of foreign demand on domestic output.

3.7 Uncovered interest parity (UIP)

$$\hat{r}_t = \hat{r}^* - \psi\hat{b}_t^* + \mathbb{E}_t\hat{r}_{t+1} - \hat{s}_t. \quad (21)$$

3.8 Net foreign assets (NFA)

The balance-of-payments identity in log-linear form:

$$\widehat{b}_{t+1}^* = \underbrace{\left(\frac{1}{\beta} - \psi\right)}_{< 1} \widehat{b}_t^* + \kappa_{EX} \left(-\theta^* \widehat{s}_t + \widehat{D}_t^*\right) - \kappa_{IM} (\widehat{p}_t^{F*} + \widetilde{y}_t), \quad (22)$$

where κ_{EX} and κ_{IM} are steady-state ratios of exports and imports to GDP. The eigenvalue $(1/\beta - \psi) < 1$ ensures stationarity.

3.9 Exchange rate pass-through to sectoral costs

Defining **import centrality**

$$\mathcal{M}_i \equiv [(\mathbf{I} - \boldsymbol{\Omega}^H)^{-1} \boldsymbol{\Omega}^F \mathbf{1}]_i, \quad (23)$$

the total exposure of sector i 's inflation to an FX/import-price shock under a peg is proportional to $\mathcal{M}_i$. Sectors with higher import centrality transmit more FX shocks to domestic prices.

3.10 Policy rules

Three alternative regimes are implemented by replacing one block in Dynare:

Regime A — Free float (benchmark):

$$\widehat{u}_t = \phi_\pi \pi_t^{DC} + \phi_y \widetilde{y}_t. \quad (24)$$

$\widehat{s}_t$ is endogenous (determined by UIP).

Regime B — Hard peg:

$$\widehat{s}_t = 0 \quad \forall t. \quad (25)$$

$\widehat{u}_t$ is residually determined by UIP (21).

Regime C — Managed float:

$$\widehat{u}_t = \phi_\pi \pi_t^{DC} + \phi_y \widetilde{y}_t + \phi_s \widehat{s}_t. \quad (26)$$

$\phi_s > 0$ implies partial exchange-rate stabilisation; $\phi_s \rightarrow \infty$ approaches a peg.

3.11 Exogenous shock processes

$$\widehat{a}_{it} = \rho_a \widehat{a}_{i,t-1} + \varepsilon_{it}^a, \quad i = 1, 2, 3, \quad (27)$$

$$\widehat{p}_t^{F*} = \rho_{pF} \widehat{p}_{t-1}^{F*} + \varepsilon_t^{pF}, \quad (28)$$

$$\widehat{D}_t^* = \rho_D \widehat{D}_{t-1}^* + \varepsilon_t^D. \quad (29)$$

3.12 Variable summary

Variable	Type	Description
$\pi_{1t}, \pi_{2t}, \pi_{3t}$	Jump	Sectoral inflation (Resource, Manuf., Services)
$\tilde{y}_t$	Jump	Output gap
$\hat{s}_t$	Jump (float) / Exog.=0 (peg)	Nominal exchange rate (dep.>0)
$\hat{i}_t$	Jump (float) / Residual (peg)	Domestic interest rate
p_{1t}, p_{2t}, p_{3t}	State (endogenous)	Sectoral price levels
$\hat{b}_t^*$	State (endogenous)	Net foreign assets
$\hat{w}_t$	Static	Nominal wage
$\pi_t^C, \hat{p}_t^C, \pi_t^{DC}$	Static	CPI inflation, CPI level, DC index
$\hat{a}_{1t}, \hat{a}_{2t}, \hat{a}_{3t}$	State (exogenous)	Sectoral TFP
$\hat{p}_t^{F*}$	State (exogenous)	Import price shock
$\hat{D}_t^*$	State (exogenous)	Foreign demand shock

4 Key Analytical Results

Lemma 1 (Output gap and markups). $(\gamma + \varphi)\tilde{y}_t = -\sum_i \lambda_{D,i}\mu_{it}$, where $\mu_{it} = p_{it} - mc_{it}$ is sector i 's log markup.

Lemma 2 (Markup–inflation link). $\mu_{it} = -\frac{1 - \hat{\delta}_i}{\hat{\delta}_i}(\pi_{it} - \beta\mathbb{E}_t\pi_{i,t+1})$.

Proposition 1 (Open-economy divine coincidence). *The DC index (19) satisfies*

$$\pi_t^{DC} = \beta \mathbb{E}_t \pi_{t+1}^{DC} + \kappa^{DC} \tilde{y}_t, \quad \kappa^{DC} = \frac{\gamma + \varphi}{\sum_k \lambda_{D,k}(1 - \hat{\delta}_k)/\hat{\delta}_k}, \quad (30)$$

with **no residual**: neither domestic TFP shocks $\hat{\mathbf{a}}_t$ nor the exchange rate $\hat{s}_t$ nor import prices $\hat{p}_t^{F*}$ appear.

Proof sketch: Embed each imported input as a fully-flexible upstream node ($\hat{\delta}_M = 1$) in an augmented network. Fully flexible nodes receive zero weight in the DC index ($w_M = 0$) because $(1 - \hat{\delta}_M)/\hat{\delta}_M = 0$. Rubbo's Proposition 1 applies verbatim to the augmented network. $\square$

Corollary (FX regime comparison). Under the float (24) with $\phi_\pi > 1$, the exchange rate absorbs import-price shocks: $\hat{s}_t \approx -\hat{p}_t^{F*}$, so all sectoral inflations converge to zero and $\tilde{y}_t \rightarrow 0$. Under the peg (25), import-price shocks pass through into domestic inflation with magnitude proportional to the import centrality $\mathcal{M}_i$ of each sector.

5 Calibration

The adjusted Calvo parameters (2) are $(\hat{\delta}_1, \hat{\delta}_2, \hat{\delta}_3) = (0.693, 0.336, 0.079)$.

6 Dynare Implementation

The model is implemented in the .mod file below. The free float is the default; switching to the peg or managed float requires uncommenting the relevant `@#define REGIME` line. Run with Dynare 5.x: `dynare open_economy_network.mod`.

Listing 1: open_economy_network.mod — full Dynare code

```

1 // =====
2 // Open Economy Production Network NK Model
3 // Rubbo (2024) extended to Small Open Economy
4 // 3 sectors: Resource (1), Manufacturing (2), Services (3)
5 //
6 // REGIMES (uncomment one):
7 // @define REGIME = "float" // Free float (default)
8 // @define REGIME = "peg" // Hard peg
9 // @define REGIME = "managed" // Managed float
10 @define REGIME = "float"
11 // =====
12
13 // -----
14 // VARIABLES
15 // -----
16 var
17     // Sectoral inflation
18     pi1 pi2 pi3
19
20     // Aggregate inflation measures
21     piDC piCPI
22
23     // Output gap and interest rate
24     y_gap i_hat
25
26     // Exchange rate (log deviation, depreciation = positive)
27     s_hat
28
29     // Sectoral price levels (log deviation)
30     p1 p2 p3
31
32     // Consumer price level (log deviation)
33     pC_hat
34
35     // Real wage (log deviation)
36     w_hat
37
38     // Net foreign assets (log deviation)
39     bstar
40
41     // TFP shocks (sectoral)
42     a1 a2 a3
43
44     // Import price shock
45     pF
46
47     // Foreign demand shock
48     D_star
49     ;
50
51 // -----
52 // EXOGENOUS SHOCKS
53 // -----
54 varexo eps_a1 eps_a2 eps_a3 eps_pF eps_D;
55
56 // -----
57 // PARAMETERS

```

```

58 // -----
59 parameters
60     // Preferences
61     BETA GAMMA VARPHI
62
63     // Adjusted Calvo parameters (Rubbo eq. after (14))
64     DHAT1 DHAT2 DHAT3
65
66     // Labour shares
67     ALPHA1 ALPHA2 ALPHA3
68
69     // Domestic input-output shares
70     OH21      // Manuf. buys from Resource (domestic)
71     OH32      // Services buys from Manuf. (domestic)
72
73     // Import shares (total per sector)
74     OF1 OF2 OF3
75
76     // Final consumption shares (domestic sectors)
77     BH1 BH2 BH3
78
79     // Total import share in final consumption
80     BF_TOT
81
82     // DC index weights
83     WDC1 WDC2 WDC3
84
85     // Open economy
86     PSI      // NFA premium
87     THETA_S  // export price elasticity
88     KAP_EX   // exports/GDP
89     KAP_IM   // imports/GDP
90     NU_D     // foreign demand sensitivity
91
92     // Policy
93     PHI_PI PHI_Y PHI_S
94
95     // Shock persistence
96     RHO_A RHO_PF RHO_D
97     ;
98
99 // -----
100 // PARAMETER VALUES
101 // -----
102
103 // Preferences
104 BETA      = 0.99;
105 GAMMA     = 1.00;
106 VARPHI    = 2.00;
107
108 // Adjusted Calvo (delta_hat from eq. (2))
109 // delta = [0.75, 0.50, 0.25], beta = 0.99
110 // dhat_i = delta_i*(1-beta*(1-delta_i))/(1-beta*delta_i*(1-delta_i))
111 DHAT1     = 0.693;
112 DHAT2     = 0.336;
113 DHAT3     = 0.079;
114
115 // Labour shares (alpha_i = 1 - sum_j(omega_H_ij + omega_F_ij))

```



```

116 ALPHA1 = 0.70;    // Resource: 1 - 0.00 - 0.30
117 ALPHA2 = 0.70;    // Manuf.: 1 - 0.20 - 0.10
118 ALPHA3 = 0.70;    // Services: 1 - 0.25 - 0.05
119
120 // Domestic I-O shares
121 OH21 = 0.20;    // Manuf. <- Resource (domestic)
122 OH32 = 0.25;    // Services <- Manuf. (domestic)
123
124 // Total import shares per sector
125 OF1 = 0.30;    // Resource
126 OF2 = 0.10;    // Manufacturing
127 OF3 = 0.05;    // Services
128
129 // Final consumption shares
130 BH1 = 0.05;
131 BH2 = 0.15;
132 BH3 = 0.80;
133 BF_TOT = 0.10;
134
135 // DC index weights w_i propto lambda_D_i*(1-dhat_i)/dhat_i
136 // lambda_D = [0.12, 0.35, 0.80] (domestic supplier centrality)
137 // raw weights:
138 // w1 = 0.12*(1-0.693)/0.693 = 0.0532
139 // w2 = 0.35*(1-0.336)/0.336 = 0.6916
140 // w3 = 0.80*(1-0.079)/0.079 = 9.328
141 // total = 10.073
142 WDC1 = 0.0053;
143 WDC2 = 0.0686;
144 WDC3 = 0.9261;
145
146 // Open economy
147 PSI = 0.020;
148 THETA_S = 2.00;
149 KAP_EX = 0.85;
150 KAP_IM = 0.45;
151 NU_D = 0.30;
152
153 // Policy
154 PHI_PI = 1.50;
155 PHI_Y = 0.50;
156 PHI_S = 0.30;    // managed float only
157
158 // Shock persistence
159 RHO_A = 0.90;
160 RHO_PF = 0.85;
161 RHO_D = 0.80;
162
163 // -----
164 // MODEL BLOCK
165 // -----
166 model(linear);
167
168 //-----
169 // [1] Sectoral Phillips curves (eqs. 3-5 in paper)
170 // pi_it = beta*(1-dhat_i)*E[pi_{i,t+1}]
171 //         + dhat_i*[alpha_i*w + omega_H_ij*p_j - p_i(-1)
172 //                   + omega_F_i*(s+pF) - a_i]
173 // Note: p_jt = p_j(-1) + pi_j is substituted inline

```

```

174 //-----
175
176 // Resource (no domestic inputs)
177 pi1 = BETA*(1-DHAT1)*pi1(+1)
178     + DHAT1*(    ALPHA1*w_hat
179               + OF1*(s_hat + pF)
180               - a1
181               - p1(-1)
182             );
183
184 // Manufacturing (buys from Resource domestic)
185 pi2 = BETA*(1-DHAT2)*pi2(+1)
186     + DHAT2*(    ALPHA2*w_hat
187               + OH21*(p1(-1) + pi1)
188               + OF2*(s_hat + pF)
189               - a2
190               - p2(-1)
191             );
192
193 // Services (buys from Manufacturing domestic)
194 pi3 = BETA*(1-DHAT3)*pi3(+1)
195     + DHAT3*(    ALPHA3*w_hat
196               + OH32*(p2(-1) + pi2)
197               + OF3*(s_hat + pF)
198               - a3
199               - p3(-1)
200             );
201
202 //-----
203 // [2] Price level laws of motion (eq. 6)
204 //-----
205 p1 = p1(-1) + pi1;
206 p2 = p2(-1) + pi2;
207 p3 = p3(-1) + pi3;
208
209 //-----
210 // [3] Consumer price level (eq. 9)
211 //-----
212 pC_hat = BH1*p1 + BH2*p2 + BH3*p3 + BF_TOT*(s_hat + pF);
213
214 //-----
215 // [4] CPI inflation (eq. 8)
216 //-----
217 piCPI = BH1*pi1 + BH2*pi2 + BH3*pi3
218       + BF_TOT*((s_hat + pF) - (s_hat(-1) + pF(-1)));
219
220 //-----
221 // [5] Real wage (eq. 7, labour supply)
222 // w_hat = pC_hat + (gamma+varphi)*y_gap
223 //-----
224 w_hat = pC_hat + (GAMMA + VARPHI)*y_gap;
225
226 //-----
227 // [6] DC inflation index (eq. 10, Proposition 1)
228 //-----
229 piDC = WDC1*pi1 + WDC2*pi2 + WDC3*pi3;
230
231 //-----

```

```

232 // [7] IS curve (eq. 11)
233 // y_gap = E[y_gap(+1)] - (1/gamma)*(i_hat - E[piCPI(+1)])
234 //       + nu_D * D_star
235 // (natural rate r_nat = 0 in deviations from SS)
236 //-----
237 y_gap = y_gap(+1) - (1/GAMMA)*(i_hat - piCPI(+1)) + NU_D*D_star;
238
239 //-----
240 // [8] UIP (eq. 12)
241 // i_hat = -psi*bstar + E[s(+1)] - s
242 //-----
243 i_hat = -PSI*bstar + s_hat(+1) - s_hat;
244
245 //-----
246 // [9] NFA accumulation (eq. 13)
247 // bstar = (1/beta - psi)*bstar(-1)
248 //       + kap_EX*(-theta_s*s + D_star)
249 //       - kap_IM*(pF + y_gap)
250 // Eigenvalue = 1/beta - psi = 0.990 < 1 (stationary)
251 //-----
252 bstar = (1/BETA - PSI)*bstar(-1)
253       + KAP_EX*(-THETA_S*s_hat + D_star)
254       - KAP_IM*(pF + y_gap);
255
256 //-----
257 // [10] Policy rule (regime-dependent)
258 //-----
259 @#if REGIME == "float"
260     // Free float: Taylor rule on DC index
261     i_hat = PHI_PI*piDC + PHI_Y*y_gap;
262
263 @#elseif REGIME == "peg"
264     // Hard peg: exchange rate fixed
265     s_hat = 0;
266     // i_hat determined residually by UIP (already in eq. 8)
267
268 @#elseif REGIME == "managed"
269     // Managed float: Taylor rule + FX stabilisation
270     i_hat = PHI_PI*piDC + PHI_Y*y_gap + PHI_S*s_hat;
271
272 @#endif
273
274 //-----
275 // [11] Shock processes (eqs. 15-17)
276 //-----
277 a1 = RHO_A*a1(-1) + eps_a1;
278 a2 = RHO_A*a2(-1) + eps_a2;
279 a3 = RHO_A*a3(-1) + eps_a3;
280 pF = RHO_PF*pF(-1) + eps_pF;
281 D_star = RHO_D*D_star(-1) + eps_D;
282
283 end;
284
285 // -----
286 // STEADY STATE
287 // (All log-deviations are zero by construction)
288 // -----
289 initval;

```

```

290     pi1 = 0; pi2 = 0; pi3 = 0;
291     piDC = 0; piCPI = 0;
292     y_gap = 0; i_hat = 0; s_hat = 0;
293     p1 = 0; p2 = 0; p3 = 0;
294     pC_hat = 0; w_hat = 0; bstar = 0;
295     a1 = 0; a2 = 0; a3 = 0;
296     pF = 0; D_star = 0;
297 end;
298
299 // -----
300 // SHOCKS (unit standard deviations)
301 // -----
302 shocks;
303     var eps_a1 = 0.01^2;    // 1% TFP shock, sector 1 (Resource)
304     var eps_a2 = 0.01^2;    // 1% TFP shock, sector 2 (Manuf.)
305     var eps_a3 = 0.01^2;    // 1% TFP shock, sector 3 (Services)
306     var eps_pF = 0.01^2;    // 1% import price shock
307     var eps_D = 0.01^2;    // 1% foreign demand shock
308 end;
309
310 // -----
311 // SOLUTION AND OUTPUT
312 // -----
313 stoch_simul(
314     order          = 1,      // linear model
315     periods        = 0,      // no simulation; compute IRFs only
316     irf            = 40,     // 40-quarter IRF horizon
317     graph_format   = pdf,
318     nograph        = 0,
319     noprint        = 0
320 );
321
322 // Print key model moments
323 disp('==_DC_index_weights_==');
324 disp([WDC1 WDC2 WDC3]);
325
326 disp('==_Import_centrality_(network_exposure_to_FX_shock)_==');
327 // M_i = [((I-OmegaH)^{-1}) * OF] -- computed externally from network
    matrices
328 // Values: [0.30, 0.16, 0.09] for Resource, Manuf., Services

```

Switching regimes. To compare the peg, change line 12 to `@define REGIME = "peg"` and re-run. For the managed float, use `"managed"`. Dynare will automatically drop or add the exchange-rate equation.

Note on the peg. Under the peg ($s_hat = 0$), the interest-rate equation i_hat is no longer set by the Taylor rule; instead it is determined residually by UIP (21) (equation [8] in the model block remains active). Dynare handles this automatically since the number of equations equals the number of variables in both cases.

Welfare comparison. After obtaining the second-order moments from `stoch_simul`, compute the per-period welfare loss:

$$\mathcal{W} = \frac{1}{2} \left[(\gamma + \varphi) \text{Var}(\tilde{y}_t) + \sum_i \lambda_{D,i} \varepsilon_i \frac{1 - \hat{\delta}_i}{\hat{\delta}_i} \text{Var}(\pi_{it}) \right]. \quad (31)$$

In Dynare add after `stoch_simul`:

```
// Welfare (Rubbo Prop. 3 adapted for open economy)
GAMMA_PHI = GAMMA + VARPHI;
EPS = 8;          // within-sector substitution elasticity (same for all
                  i)
LD = [0.12, 0.35, 0.80]; // lambda_D
DHAT = [DHAT1, DHAT2, DHAT3];

W_output = 0.5 * GAMMA_PHI * oo_.var(strmatch('y_gap', M_.endo_names,
'exact'), ...
    strmatch('y_gap', M_.endo_names, 'exact'));

W_pi = 0;
for i = 1:3
    var_name = ['pi', num2str(i)];
    idx = strmatch(var_name, M_.endo_names, 'exact');
    W_pi = W_pi + 0.5 * LD(i) * EPS * (1-DHAT(i))/DHAT(i) * oo_.var(
        idx, idx);
end

W_total = W_output + W_pi;
fprintf('Welfare_loss(x10^4): %.4f\n', W_total * 1e4);
fprintf('Output_gap_component: %.4f\n', W_output * 1e4);
fprintf('Inflation_component: %.4f\n', W_pi * 1e4);
```

References

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Table 1: Calibration — 3-sector oil-exporter (quarterly)

Parameter	Value	Description	Source
<i>Preferences</i>			
β	0.99	Discount factor	Standard
γ	1.00	Risk aversion / inverse EIS	Standard
φ	2.00	Inverse Frisch elasticity	?
η	1.50	Home-foreign substitution	?
ω	0.70	Home bias in consumption	Data average
<i>Domestic input-output matrix Ω^H</i>			
ω_{21}^H	0.20	Manuf. buys 20% from Resource	BEA I-O
ω_{32}^H	0.25	Services buys 25% from Manuf.	BEA I-O
<i>Imported input matrix Ω^F</i>			
ω_1^F	0.30	Resource: 30% imported machinery	SOE data
ω_2^F	0.10	Manuf.: 10% imported components	SOE data
ω_3^F	0.05	Services: 5% indirect imports	SOE data
<i>Labour shares $\alpha_i = 1 - \sum_j (\omega_{ij}^H + \omega_{ij}^F)$</i>			
α_1	0.70	Resource	Derived
α_2	0.70	Manufacturing	Derived
α_3	0.70	Services	Derived
<i>Final consumption shares</i>			
$\beta_1^H, \beta_2^H, \beta_3^H$	0.05, 0.15, 0.80	Domestic sectors	Expenditure survey
$\beta^{F,\text{tot}}$	0.10	Total import share in consumption	Trade data
<i>Pricing (Calvo)</i>			
δ_1	0.75	Resource: flexible (commodity prices)	?
δ_2	0.50	Manufacturing: intermediate	?
δ_3	0.25	Services: sticky	?
ε_i	8.00	Within-sector substitution elasticity	Standard NK
<i>Open economy</i>			
ψ	0.020	NFA debt-elastic premium (ensures stationarity)	?
θ^*	2.00	Export price elasticity	?
κ_{EX}	0.85	Exports/GDP ratio	Oil exporter data
κ_{IM}	0.45	Imports/GDP ratio	Oil exporter data
ν_D	0.30	Foreign demand sensitivity	Calibrated
<i>Monetary policy</i>			
ϕ_π	1.50	Taylor rule: inflation	?
ϕ_y	0.50	Taylor rule: output gap	?
ϕ_s	0.30	Managed float: FX weight	Calibrated
<i>Shock persistence</i>			
ρ_a	0.90	TFP	Standard
ρ_{pF}	0.85	Import price	Commodity price
ρ_D	0.80	Foreign demand	Standard
<i>Derived network objects</i>			
$\lambda_{D,1}, \lambda_{D,2}, \lambda_{D,3}$	0.12, 0.35, 0.80	Domestic supplier centrality	Eq. (1)
$w_1^{DC}, w_2^{DC}, w_3^{DC}$	0.005, 0.069, 0.926	DC index weights	Eq. (19)
κ^{DC}	0.298	DC Phillips curve slope	Eq. (1)
$\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$	0.30, 0.16, 0.09	Import centrality	Eq. (23)